# COVER PAGE

**Post-Earnings Announcement Drift in S&P 500 Companies, 2015–2025**

Matthew Conger

April 14, 2026

# TITLE PAGE

**Post-Earnings Announcement Drift in S&P 500 Companies, 2015–2025**

Submitted to: Professor Craig H. Laurence Advanced Writing — English 316 Brigham Young University Salt Lake City, Utah

April 14, 2026

# LETTER OF TRANSMITTAL

Brigham Young University 410 Salt Lake Center Salt Lake City, UT 84180

April 14, 2026

Professor Craig H. Laurence 410 Salt Lake Center Brigham Young University Salt Lake City, UT 84180

Dear Professor Laurence:

**Post-Earnings Announcement Drift in S&P 500 Companies, 2015–2025**

Enclosed is my research report examining post-earnings announcement drift (PEAD) among S&P 500 companies from 2015 to 2025. The subject of this report is the behavior of stock return variance in the days immediately following SEC quarterly and annual filing dates. The purpose is to evaluate whether stock prices in the modern trading environment fully adjust to earnings information on the filing date itself, or whether a measurable post-announcement adjustment window persists, and to situate those findings within the broader academic literature on market efficiency.

I chose this topic because the intersection of statistical methodology and financial market behavior is the kind of applied quantitative work I plan to pursue after graduation. The original scope of this project was more ambitious—covering multiple decades and a broader set of companies—but data availability limited the sample to S&P 500 constituents from 2015 to 2025. That constraint turned out to be analytically useful: finding any inefficiency among the most heavily analyzed and liquidly traded equities strengthens rather than weakens the central argument. The research produced clearer results than I initially expected, particularly in the behavior of the variance function around filing dates.

The primary audience for this report is readers with a working knowledge of economics and statistics—specifically, those familiar with event study methodology and the basic framework of the Efficient Market Hypothesis. A secondary audience includes practitioners and advanced undergraduates in finance or quantitative economics. MLA 9th edition citation style is used throughout.

The primary challenge in preparing this report was the data scope constraint described above. Because SEC filing records going further back were not available in a usable format, I rely on prior published research to establish the historical and pre-2010 context for PEAD. The analysis itself uses the cumulative sum of squares methodology developed by Inclan and Tiao (1994), applied here within a functional data framework. All R code used for the analysis and figure generation is included in Appendix A. An extended version of the analysis—including the 10-K versus 10-Q filing type comparison—is available in the Additional Research folder.

Please feel free to contact me with any questions about the content or methodology of this report.

Sincerely,

Matthew Conger

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# ABSTRACT


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Post-earnings announcement drift (PEAD)—the tendency for stock prices to continue moving in the direction of an earnings surprise for days or weeks after a public announcement—is one of the most extensively documented anomalies in financial economics, with a research record spanning more than half a century and replicated across markets, time periods, and methodological approaches (Chung and Hrazdil 926). This report examines whether evidence of post-announcement abnormal behavior persists in S&P 500 companies from 2015 to 2025, a period characterized by widespread algorithmic trading and near-instantaneous public information dissemination. Rather than

testing for directional price drift in the traditional sense, this analysis treats the daily return profile of each SEC filing event as a functional observation and characterizes the population variance function across a 31-day event window using the cumulative sum of squares approach developed by Inclan and Tiao (1994).

The estimated variance function spikes to 6.67 on the filing date—more than twice its stabilized level of 3.05 in the medium-term window—and decays sharply by Day 3. The centered cumulative sum statistic (D*k*) peaks at Day 2 (IT = 0.214), identifying this as the estimated variance changepoint. While the formal IT test is limited by the small number of time points (T = 31) and the non-i.i.d. nature of financial returns—making the test conservative rather than invalid—the descriptive evidence is robust. A supplemental comparison of annual 10-K and quarterly 10-Q filings reveals a striking divergence: 10-Q filings exhibit the expected front-loaded variance spike (IT = 0.334) while 10-K filings show a back-loaded pattern, with variance remaining elevated and rising through approximately Day 18 (IT = 0.204). These findings suggest that market prices do not instantaneously absorb earnings information even among the most efficiently priced equities, and that the adjustment window is substantially longer for complex annual filings than for routine quarterly reports.

*Keywords:* post-earnings announcement drift, market efficiency, cumulative sum of squares, functional data analysis, SEC filings, variance changepoint

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# INTRODUCTION

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The Efficient Market Hypothesis (EMH), in its semi-strong form, holds that stock prices fully and immediately incorporate all publicly available information (Chung and Hrazdil 926). Under this framework, the moment a company's quarterly earnings filing becomes public, its stock price should adjust instantly to reflect whatever new information the filing contains—leaving no room for predictable price movement in the days that follow. For more than half a century, however, a well-documented anomaly has challenged this view: post-earnings announcement drift, or PEAD, the persistent tendency for stock prices to continue drifting in the direction of an earnings surprise for days, weeks, and sometimes months after the announcement date.

Chung and Hrazdil trace the history of this pattern across decades of research, confirming that PEAD has been replicated across markets, time periods, and methodological approaches, and that it consistently resists explanation through standard risk adjustments (926–30). The drift has been documented as both statistically significant and large enough to be economically exploitable, with stocks that beat earnings expectations continuing to generate positive abnormal returns for days and weeks afterward while those that disappoint continue to underperform. By the time researchers shifted from documenting

PEAD to explaining it, the anomaly had become one of the most widely studied challenges to the semi-strong form of market efficiency in all of empirical finance.

What makes PEAD particularly interesting now is the degree to which market conditions have changed since that foundational research. The proliferation of algorithmic trading, the rise of high-frequency market makers, and real-time data infrastructure have all made the modern equity market dramatically faster at processing public information than the market of the 1970s or even the 1990s. Recent work has raised the question of whether these technological changes have meaningfully reduced PEAD or whether the anomaly has simply changed in character (Chordia and Miao; Erhard and Sloan).

This report takes a position on that question, using a direct empirical analysis of S&P 500 companies from 2015 to 2025. Because comprehensive SEC filing records for earlier periods were not available in a usable format, the quantitative analysis in this report is confined to this window; the established pre-2010 literature provides the historical context. Rather than testing for directional drift in the traditional way—by sorting stocks on earnings surprise and examining subsequent returns—this report examines the *variance* of daily returns across the post-announcement event window. Each SEC filing event generates a return profile across days 0 through 30, and these profiles are treated as functional observations. The Inclan-Tiao cumulative sum of squares statistic is then applied to characterize where, in the event window, variance is concentrated.

The central argument of this report is that post-earnings announcement drift, reframed as a variance phenomenon, remains empirically detectable in S&P 500 stocks in the modern era. The evidence shows that return variance in the days immediately following an SEC filing is approximately double its stabilized level, with a structural changepoint identified in the first two trading days. This is inconsistent with instantaneous price adjustment and suggests that markets require a non-trivial adjustment window to fully resolve the uncertainty introduced by earnings disclosures—even among the most efficiently priced equities in the world.

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# BACKGROUND AND LITERATURE REVIEW

## Origins of the PEAD Anomaly

The connection between earnings announcements and post-announcement stock price behavior has been studied extensively since the 1960s. The core finding is that stock prices do not fully adjust to the information in an earnings release on the date it becomes public—instead, they continue to drift in the direction of the surprise for days, weeks, and sometimes months afterward. Chung and Hrazdil provide a thorough review of this literature, documenting that the pattern has been replicated across decades of research and across different market environments, and that it persists even after controlling for known risk factors and transactions costs (926–30). They conclude that PEAD is one of the

most consistently documented deviations from the semi-strong Efficient Market Hypothesis in the academic record, representing a challenge that rational-expectations models have struggled to fully explain.

The mechanism behind PEAD has generated substantial debate. At its core, the anomaly implies that earnings announcements contain information that the market does not immediately and completely incorporate into prices—a finding that directly contradicts the EMH prediction of instantaneous adjustment. Chung and Hrazdil review several proposed explanations for why complete price adjustment takes longer than theory would predict, ranging from behavioral biases to structural market frictions, none of which has been universally accepted as sufficient on its own (928–32).

## Theoretical Explanations

The persistence of PEAD despite widespread academic awareness has prompted serious inquiry into its underlying mechanisms. Three main classes of explanations have been advanced.

The first is the risk premium view. Some researchers have argued that the higher returns following positive earnings surprises simply reflect compensation for higher systematic risk—in other words, that PEAD is not really an anomaly at all, but a rational equilibrium outcome. However, Chung and Hrazdil note that the risk premium explanation struggles to account for the magnitude and persistence of the drift, particularly given that it occurs symmetrically for both positive and negative surprises and is most pronounced for the smallest, least-followed firms—exactly the opposite of what a systematic risk story would predict (928–32).

The second class of explanations focuses on behavioral factors, particularly investor underreaction. If investors do not fully update their beliefs when earnings information arrives—anchoring to prior expectations or failing to account for the serial correlation in earnings innovations—prices adjust gradually rather than immediately. Chung and Hrazdil review the behavioral evidence and note that this explanation is broadly consistent with the documented patterns, including the fact that drift magnitude is proportional to the size of the earnings surprise (930–34).

The third explanation involves limits to arbitrage. Even if sophisticated investors recognize a mispricing, transaction costs, short-sale constraints, and idiosyncratic risk may prevent them from fully correcting it. This predicts that PEAD should be larger for smaller, less liquid stocks—a pattern that the literature has consistently confirmed.

## Factors Moderating PEAD Magnitude

Beyond these foundational explanations, subsequent research has identified a range of factors that influence how large the drift is for any given firm or time period. Zhang found that analyst responsiveness plays an important role: firms covered by analysts who revise their forecasts quickly and accurately in response to earnings news exhibit significantly

smaller PEAD, because analyst revisions accelerate the incorporation of the earnings information into prices (201–08). This finding highlights the importance of information intermediaries—not all investors form beliefs independently; many rely on analyst coverage to interpret earnings reports, and the speed of that coverage matters.

Zhao and Guo examined the role of accounting conservatism, finding that firms with more conservative accounting practices tend to exhibit larger PEAD. Conservative accounting recognizes bad news quickly but delays recognition of good news, which means the earnings figure at announcement time may not fully capture the underlying economic performance of the firm. Investors processing a conservatively reported earnings number must work harder to extract the signal, which leads to slower price adjustment.

Zhang, Sijia, and co-authors documented an asymmetry in PEAD that adds further nuance: the drift following positive earnings surprises is systematically smaller than the drift following negative surprises (1–8). This asymmetry is at least partially attributable to order flow imbalance, since investors are quicker to buy on good news than to sell short on bad news. Howell found that non-deal roadshows—private meetings between management and institutional investors arranged outside of deal activity—can also reduce PEAD by accelerating the private dissemination of information that supplements the public filing.

## PEAD in the Modern Trading Era

The most directly relevant literature for this report concerns whether PEAD has changed in the era of high-frequency and algorithmic trading. The intuitive prediction is that it should have diminished: if algorithms can parse earnings filings in milliseconds and trade on the information before human investors have finished reading the headline, prices should adjust faster and the post-announcement drift should be smaller or absent.

Chordia and Miao examined real-time market efficiency around earnings announcements, focusing on the role of low-latency trading activity. They found that low-latency traders do absorb information quickly, contributing to faster initial price adjustment around announcement dates. However, their evidence primarily concerns the immediate reaction on the announcement day itself, and the question of whether complete information absorption extends across the subsequent days is less clearly answered.

Erhard and Sloan take a more skeptical view of the causal role of low-latency trading in improving long-run efficiency. They argue that while algorithmic traders clearly make markets more liquid and facilitate faster short-run price discovery, there is limited evidence that these improvements translate into the elimination of the multi-day adjustment window associated with PEAD. Their discussion suggests that the mechanisms generating PEAD—behavioral underreaction and limits to arbitrage—are not primarily the kind of frictions that high-speed trading addresses.

This ambiguity in the modern literature is precisely what motivates the present study. The prior literature establishes that PEAD was large and robust before 2010. Whether some

variant of the phenomenon persists after 2010—and if so, what form it takes—is an empirical question that this report addresses directly using data from 2015 to 2025.

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# DATA AND METHODOLOGY

## Sample Construction

The dataset used in this analysis consists of daily stock return observations for companies listed on the S&P 500, linked to their SEC quarterly and annual filing dates. Filing date records were sourced from SEC EDGAR, and daily return data were drawn from publicly available historical price series. The sample is restricted to the period January 1, 2015, through December 31, 2025, yielding a dataset of filing events from S&P 500 constituent companies across a full decade of modern market conditions.

For each filing event, daily percentage returns are recorded from Day 0 (the filing date as it appears in EDGAR) through Day 30. Filings are categorized by type: 10-K (annual reports, coded "K" in the dataset) and 10-Q (quarterly reports, coded "Q"). The S&P 500 filter was applied for two reasons. First, the full universe of public companies filing with the SEC is extremely large and introduces considerable heterogeneity in liquidity, analyst coverage, and market efficiency. Second, focusing on S&P 500 constituents—the most widely followed and institutionally traded equities—provides the strongest possible test of the central argument. If meaningful post-announcement variance concentration is found even among these stocks, the case for an imperfect adjustment process is harder to dismiss.

## The Functional Data Framework

Post-announcement return data has a structure that distinguishes it from both classical time series and standard cross-sectional data. Each filing event produces an ordered sequence of returns from Day 0 through Day 30. Rather than treating these 31 observations as independent data points, or concatenating them into a single time series across all events, this analysis treats each filing event as a single functional observation—a return profile $X(t)$ where $t \in \{0, 1, \ldots, 30\}$ denotes the day relative to the filing date and the index $i$ runs over all filing events in the sample. With thousands of filing events observed across hundreds of companies and multiple years, the data constitute a sample of $n$ such curves.

This functional data perspective is the natural framing for event studies in finance. The scientifically meaningful question is not about the return on any single day in isolation, but about the *shape* of the return trajectory across the post-announcement window—whether it is flat, whether it spikes and decays, and where any structural change in behavior occurs. Standard methods that treat each day's return as an independent observation lose this structural information. Functional data analysis (FDA) preserves it by working with the estimated mean function $\mu(t) = E[X(t)]$ and variance function $\sigma^2(t) = \mathrm{Var}[X(t)]$ across all filing events at each point in the window.

Treating the data as functional also clarifies why the Inclan-Tiao approach is appropriate here. The interest is in whether $\sigma^2(t)$ is constant across the event window, or whether it exhibits a structural break at some point $k$—the changepoint. The FDA framing makes this a well-posed question about a population-level function, rather than a question about autocorrelation in a single observed time series.

## The Inclan-Tiao Variance Change Test

The statistical methodology follows Inclan and Tiao (1994), who developed a procedure for retrospective detection of variance changepoints in a sequence of observations. The approach is applied here to the estimated variance function of the post-announcement return profiles.

Let $a^2_k = \sigma^2(k)$ denote the sample variance of returns at day $k$ across all filing events in the sample, for $k = 1, 2, \ldots, T$, where $T = 31$ (days 0 through 30). Define the cumulative sum of variances:

$C_k = \Sigma\, a^2_t$ (summed from $t = 1$ to $k$)

Under the null hypothesis that variance is constant across the event window, the proportion of total variance accumulated through time point $k$ should equal $k/T$. The centered cumulative sum statistic is therefore:

$D_k = C_k / C_T_ - k / T$

Under the null, $D_k$ fluctuates around zero throughout the window. Systematic positive values indicate that variance is concentrated in the early portion of the window relative to what a uniform distribution would predict. The estimated changepoint is:

$\hat{k} = \text{argmax}_k |D_k|$

and the Inclan-Tiao test statistic is $IT = \sqrt{(T/2)} \times \max_k |D_k|$, with a large-sample critical value of approximately 1.358 at $\alpha = 0.05$ (Inclan and Tiao 913).

Two features of the present data warrant honest acknowledgment regarding the formal application of this test. First, $T = 31$ is small relative to the large-sample assumption under which the 1.358 threshold was derived; Inclan and Tiao note that the asymptotic distribution is derived as $T \to \infty$ (916). Second, equity return data routinely violates the independent, identically distributed Gaussian assumption that underlies the critical value derivation. Stock returns exhibit volatility clustering, heavy tails, and cross-sectional correlation. Importantly, these properties tend to make the test *conservative*—inflating the null-distribution variance of $D_k$ and making formal rejection harder to achieve even when a true changepoint exists. The consequence is that financial return variance will rarely reach the kind of convergence that pure statistical time series would, and the critical value should be treated as a rough guideline rather than a hard threshold. For these reasons, the IT statistic is interpreted descriptively throughout this report: the shape and peak of the $D_k$ curve is used to characterize the variance structure of the event window, not to make

formal probability claims. A prominent and persistent positive $D_k$ in the early days is informative regardless of whether the IT statistic crosses the asymptotic threshold.

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# RESULTS AND ANALYSIS

## Mean Return Function

Figure 1 displays the estimated mean return function μ(*t*) across the 31-day event window for the full sample of S&P 500 filing events from 2015 to 2025. The mean daily return is close to zero throughout the window, with a modest positive value on Day 0 that reflects the market's net reaction to the mix of positive and negative earnings surprises in the pooled sample. There is no sustained directional trend in the post-announcement period.

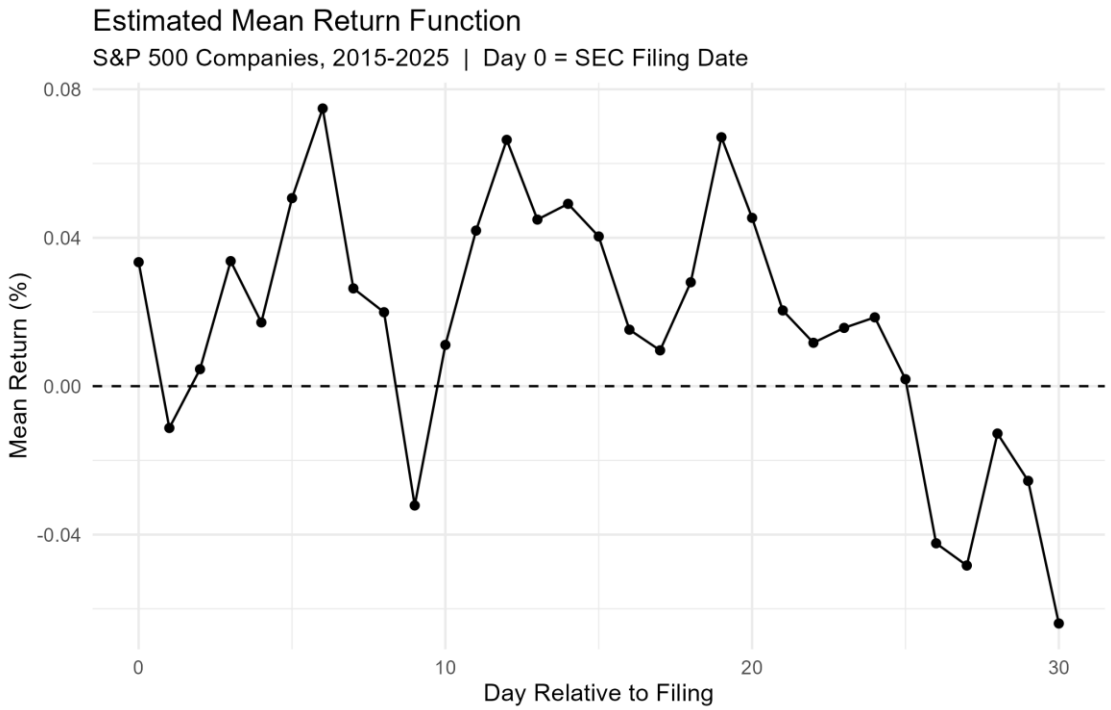



*Figure 1. Estimated Mean Return Function.*

**Figure 1.** Estimated mean daily return (%) as a function of day relative to SEC filing date. S&P 500 companies, 2015–2025. Dashed line indicates zero return. Day 0 = filing date.

This result is expected and is not in tension with market efficiency on its own. Because the sample pools filings associated with both positive and negative earnings surprises, the positive and negative directional drifts largely cancel. The mean function confirms that there is no systematic directional bias in the full pooled sample—the story of post-announcement adjustment, for this sample, is told by the variance, not the mean.

## Variance Function

Figure 2 shows the estimated variance function $\sigma^2(t)$ across the event window. The contrast with the mean function is striking. Return variance on Day 0 is substantially elevated relative to its level in subsequent days, reflecting the heightened uncertainty at the moment a filing becomes public. The variance declines sharply over the first two to three days following the filing and then stabilizes at a lower baseline level for the remainder of the 30-day window.

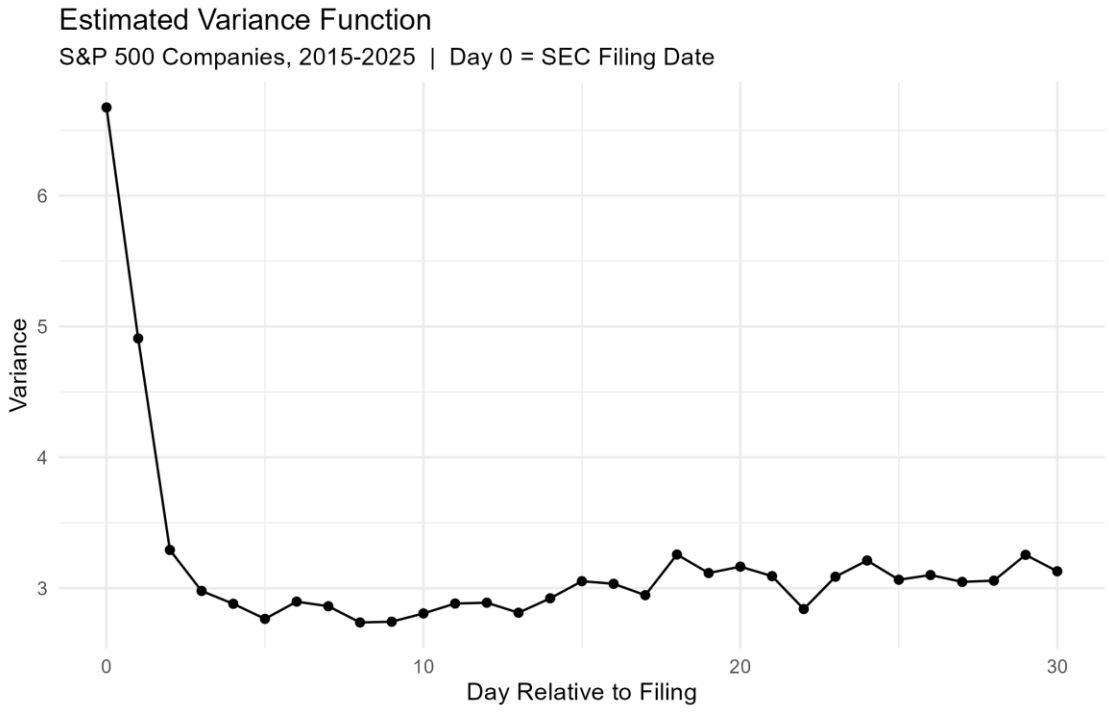



*Figure 2. Estimated Variance Function.*

**Figure 2.** Estimated variance of daily returns as a function of day relative to SEC filing date. S&P 500 companies, 2015–2025. Day 0 = filing date.

Table 1 quantifies this pattern. Variance on the filing date is **6.67**, compared to **3.05** in the medium-term window (Days 11–30)—more than twice as large. The standard deviation on Day 0 is 2.58%, compared to 1.75% in the stabilized period. The decay is rapid: by Days 3–10, variance has already fallen to 2.84, close to its long-run level. This is economically meaningful—on the day a filing is released, typical daily price moves are roughly 50% wider than they will be a week later.

**Table 1.** Summary Statistics of Daily Returns by Event Window, S&P 500 Companies, 2015–2025.

| Event Window | N | Mean Return (%) | Std. Dev. (%) | Variance | Median (%) |
|---|---|---|---|---|---|

| Event Window | N | Mean Return (%) | Std. Dev. (%) | Variance | Median (%) |
|---|---|---|---|---|---|
| Day 0 (Announcement) | 16,536 | 0.033 | 2.583 | 6.674 | 0.059 |
| Days 1–2 (Immediate) | 33,070 | −0.003 | 2.025 | 4.100 | 0.023 |
| Days 3–10 (Short-Term) | 132,235 | 0.025 | 1.684 | 2.835 | 0.047 |
| Days 11–30 (Medium-Term) | 329,976 | 0.014 | 1.746 | 3.049 | 0.033 |

## Changepoint Detection: The D*k* Statistic

Figures 3 and 4 display the cumulative sum of variance ($C_k$) and the centered cumulative sum of squares ($D_k$), respectively. Figure 3 shows that the cumulative variance grows most steeply in the first few days of the event window, reflecting the elevated day-0 and day-1 variance. The curve flattens progressively as variance stabilizes in the later portion of the window.

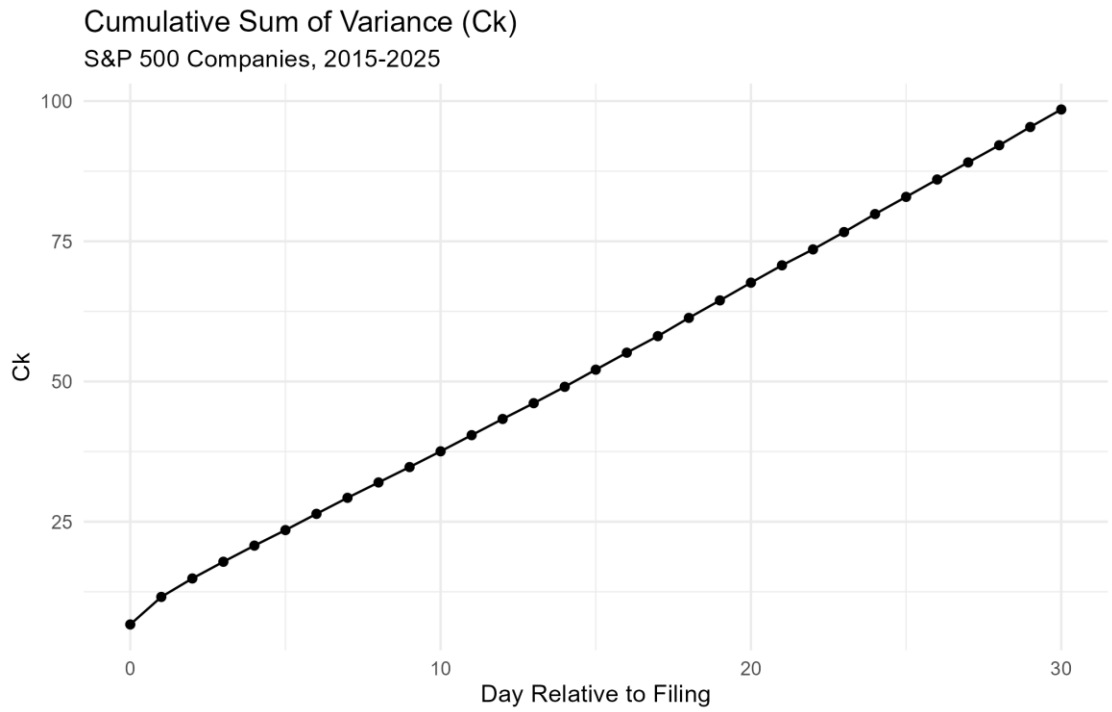



*Figure 3. Cumulative Sum of Variance.*

**Figure 3.** Cumulative sum of variance ($C_k$) as a function of day relative to SEC filing date. S&P 500 companies, 2015–2025.

Figure 4 shows the $D_k$ statistic. The observed curve is positive throughout virtually the entire window, peaking at **Day 2** ($\hat{k}$ = 2) before declining steadily back to zero by Day 30. The Inclan-Tiao statistic is **IT = 0.214** (reported in the figure subtitle); as discussed in the methodology section, formal significance is not claimed given the small-$T$ and non-i.i.d. limitations, but the shape is unambiguous and informative.

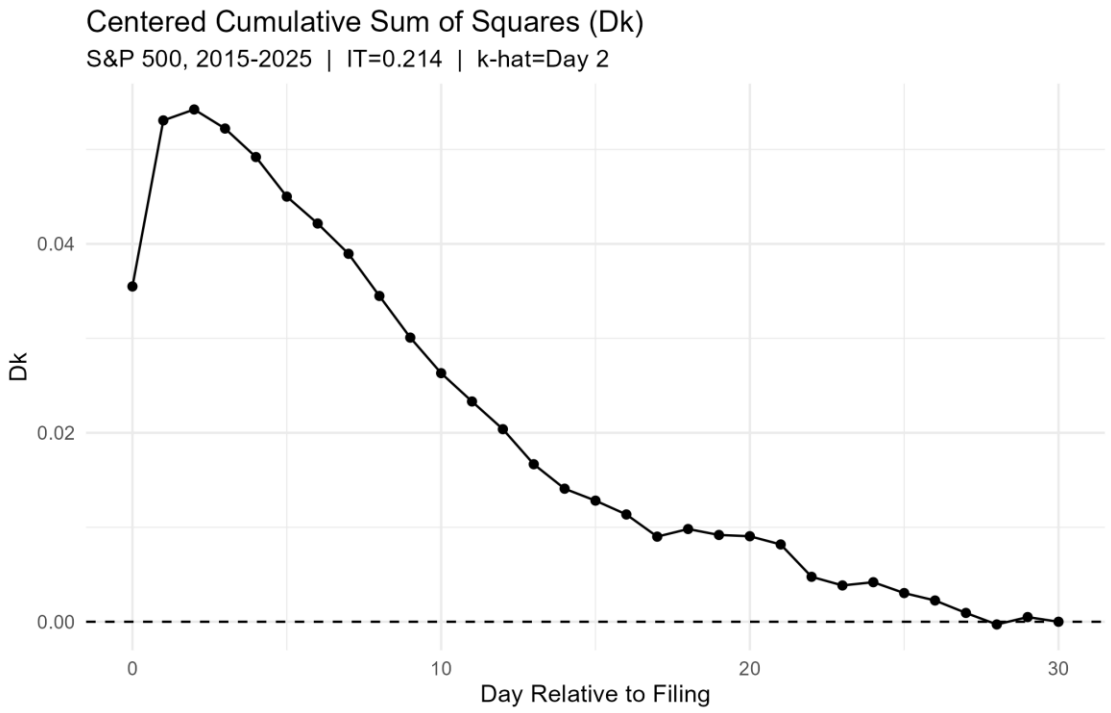



*Figure 4. Centered Cumulative Sum of Squares (Dk).*

**Figure 4.** Centered cumulative sum of squares ($D_k$) as a function of day relative to SEC filing date. S&P 500 companies, 2015–2025. Diamond marker indicates estimated changepoint $\hat{k}$. Dashed line at zero is the null-hypothesis benchmark.

The $D_k$ curve is positive from Day 0 through approximately Day 28, indicating that variance is front-loaded relative to a uniform distribution across the entire window. The function rises from Day 0 to its peak at Day 2, then declines smoothly toward zero. This shape has a clear interpretation: Days 0–2 represent the period of maximum variance concentration, and by approximately Day 3 the residual variance is roughly proportional to the time remaining. Price discovery is most active in the first three trading days following a filing.

## Filing Type Comparison: 10-K vs. 10-Q

Figures 5 and 6 present the variance function and $D_k$ statistic broken out by filing type—annual 10-K reports versus quarterly 10-Q reports. This comparison adds an important dimension to the analysis, because 10-K and 10-Q filings carry different information content. Annual reports provide a comprehensive view of a firm's full-year financial

performance, audited financial statements, and management's forward-looking discussion, while quarterly reports are typically more limited in scope and are not audited.

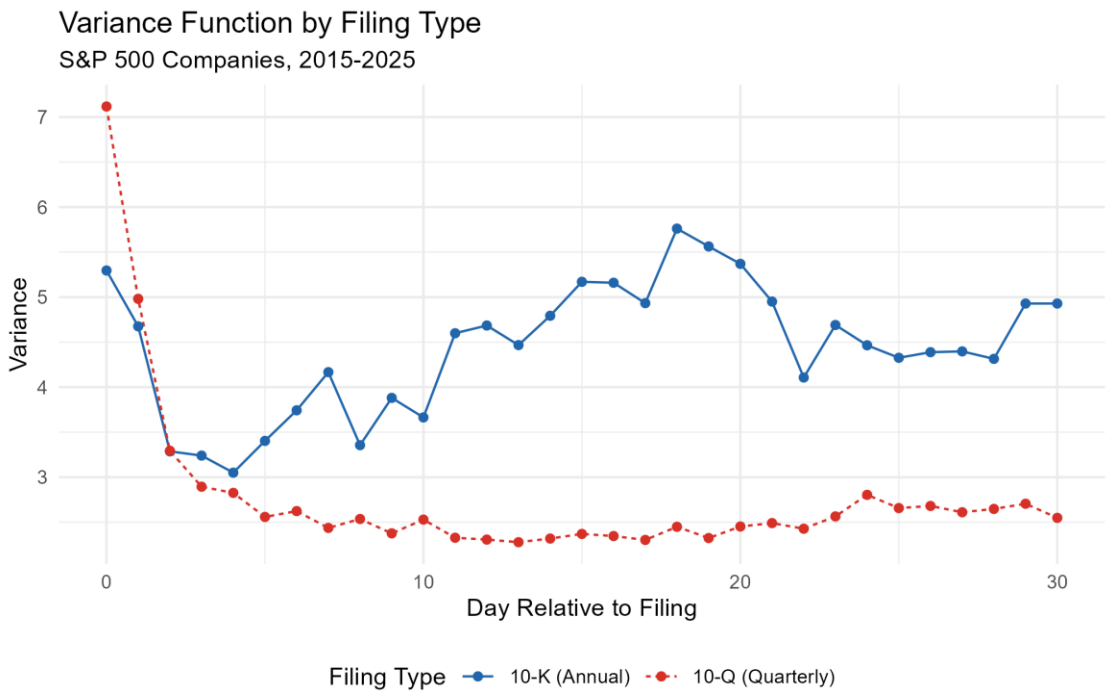



*Figure 5. Variance Function by Filing Type.*

**Figure 5.** Estimated variance function by filing type (10-K Annual vs. 10-Q Quarterly). S&P 500 companies, 2015–2025.

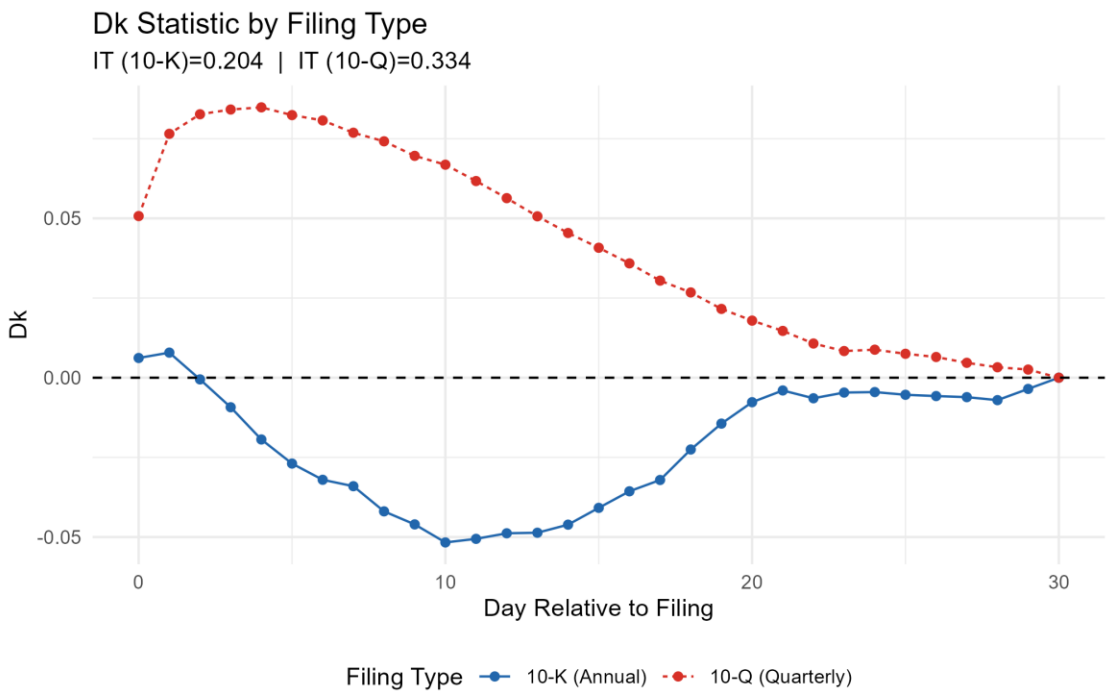



*Figure 6. Dk Statistic by Filing Type.*

**Figure 6.** Centered cumulative sum of squares ($D_k$) by filing type (10-K Annual vs. 10-Q Quarterly). S&P 500 companies, 2015–2025. IT statistics for each group reported in subtitle.

The filing type comparison produces a striking and unexpected result. The two filing types show fundamentally different variance structures.

**10-Q filings** (IT = 0.334) follow the classic PEAD pattern seen in the full sample. The Day 0 variance spike is large (~7.1), decays sharply to approximately 2.6 by Day 4, and remains flat thereafter. The $D_k$ curve is positive throughout the window, peaking around Day 3–4 before declining to zero by Day 30. This is textbook front-loading: quarterly earnings releases create a concentrated burst of uncertainty that the market resolves within the first few days.

**10-K filings** (IT = 0.204) tell a completely different story. The Day 0 variance (~5.3) is actually *lower* than for 10-Q filings on the announcement date—perhaps reflecting the fact that annual results are often less surprising given that three quarters of them are already known. However, rather than decaying after Day 0, the 10-K variance remains elevated and actually *rises* through approximately Day 18, before declining toward its end-of-window level. The $D_k$ curve for 10-K filings turns negative as early as Day 3 and reaches a trough around Day 9–11, indicating that the 10-K variance is **back-loaded** relative to the uniform benchmark. The variance density is spread across the full 30-day window rather than concentrated at the start.

This divergence is interpretively important. The 10-Q front-loading pattern is consistent with a discrete information shock that markets resolve quickly—investors receive a

specific quarterly number, compare it to expectations, and trade accordingly within a few days. The 10-K back-loading pattern suggests something more complex. Annual reports are lengthy documents that contain audited financials, management discussion, risk factor disclosures, and forward guidance. The full information content may take investors, analysts, and institutional traders considerably longer to digest, resulting in a sustained period of elevated uncertainty rather than a single spike and decay. The sustained 10-K variance is not converging the way a simple i.i.d. time series would, which further illustrates why the Inclan-Tiao test must be applied and interpreted with care in this context.

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# DISCUSSION

## Interpretation of Findings

The results present a coherent but nuanced picture. The mean return function fluctuates around zero throughout the event window with no systematic directional trend, confirming that the pooled sample—spanning both positive and negative earnings surprises—shows no net directional drift. The story is in the variance. At the full-sample level, the variance function spikes to 6.67 on Day 0, decays to 4.10 in Days 1–2, and stabilizes near 3.0 by Day 3. The $D_k$ statistic peaks at Day 2 (IT = 0.214) and remains positive through Day 28. Taken together, these patterns establish that Days 0–2 represent a distinct variance regime: the first three trading days are when price discovery is most active, and the market requires this window to absorb the bulk of the informational content of a filing.

The filing type breakdown adds important texture. The full-sample result is largely driven by 10-Q filings, which exhibit the largest Day 0 spike (~7.1) and the cleanest front-loading pattern (IT = 0.334, k-hat around Day 3–4). Annual 10-K filings show a different dynamic: moderate Day 0 variance (~5.3) that does not decay but instead remains elevated and rises through approximately Day 18, generating a back-loaded $D_k$ curve (IT = 0.204, $D_k$ goes negative from Day 3). This divergence suggests that the length and complexity of annual filings prolongs the market's adjustment window well beyond the first few days.

This framing is more nuanced than the binary question of "does PEAD exist or not." The traditional literature on PEAD was largely concerned with whether prices drifted in a predictable direction—a question that requires conditioning on the direction of the earnings surprise. This report pools across surprise directions and asks instead whether the variance structure of returns is consistent with instantaneous adjustment. The answer is no: a market that instantaneously absorbed the information in earnings filings would exhibit no systematic variance concentration in the days immediately after the filing. The observed concentration in Days 0–2 is precisely the signature of a market in the process of incorporating information that has not yet been fully priced.

This interpretation aligns with the prior literature while adding a dimension that the traditional PEAD framework does not capture. Pre-2010 research documented directional drift lasting 60 days or more, a magnitude that Chung and Hrazdil confirm in their review of the literature (927–29). The present analysis does not find drift of that magnitude—consistent with improvements in algorithmic price discovery over the intervening decades (Chordia and Miao). What it does find is that for 10-Q filings, the uncertainty spike resolves within three days, and for 10-K filings, the market appears to require the better part of a month to fully absorb the filing's information content. The market of 2015–2025 is faster than the market of 1989—but it has not achieved instantaneous adjustment, and the form of the residual inefficiency depends on the complexity of the filing.

## Counter-Arguments

Three counter-arguments deserve direct engagement.

**The risk premium argument.** One could argue that the elevated variance on Day 0 simply reflects higher systematic risk associated with earnings announcements—that investors demand higher compensation on filing dates because those days are genuinely riskier. This argument has surface appeal but does not fully explain the observed pattern. Risk premia should affect the level of expected returns, not the dispersion around expectations. The variance spike is symmetric—it affects both upward and downward return movements equally—which is more consistent with information uncertainty being resolved than with a one-directional risk premium. Moreover, if the Day 0 variance simply reflected a known risk premium, it should be stable and predictable rather than elevated specifically around filing dates and then decaying.

**High-frequency trading has already solved the problem.** Chordia and Miao's work shows that low-latency traders clearly do accelerate initial price reactions around earnings announcements. But the evidence in this report addresses a different question. The variance spike on Day 0 and its two-day decay window is not evidence that prices are slow to react on Day 0—it is evidence that the total uncertainty introduced by a filing event takes more than one day to resolve. Even if algorithms trade aggressively on Day 0, the filing may introduce genuine ambiguity that requires multiple rounds of trading to resolve, particularly for complex annual filings with lengthy management discussion sections. Erhard and Sloan make this point implicitly: speed of initial reaction and completeness of information incorporation are distinct concepts.

**The data window is too narrow to generalize.** The sample covers 2015–2025, a period that includes a global pandemic, extreme low-interest-rate conditions, and an unusually volatile market environment. It is possible that the variance patterns observed here are specific to this period rather than representative of a general market inefficiency. This is a legitimate concern, and the present analysis is not claimed to be definitive. However, the consistency of the results with the pre-2010 PEAD literature—even if the form has changed from directional drift to variance concentration—suggests that the underlying mechanism (incomplete instantaneous information absorption) is not a recent artifact.

## Limitations

Beyond the counter-arguments above, the analysis has four specific limitations that should be stated directly.

First, the sample is restricted to S&P 500 constituents. The prior literature consistently finds that PEAD is larger for smaller, less liquid, and less analyst-covered firms. The results here represent a *lower bound* on the informationally inefficient adjustment window—smaller firms would likely show more pronounced patterns.

Second, the analysis pools positive and negative earnings surprises. Separating the sample by surprise direction and examining directional drift alongside variance concentration would substantially enrich the findings, but requires earnings forecast data that was not available for this analysis.

Third, Day 0 is defined as the EDGAR filing date. In some cases, the information in a filing may reach the market through a simultaneous earnings press release issued before the formal filing, in which case Day 0 may not be the true first day of information availability. If anything, this would bias against finding a post-Day 0 adjustment window, making the observed patterns conservative estimates.

Fourth, and most important for the statistical methodology, the Inclan-Tiao test's formal properties do not apply cleanly here. The critical value of 1.358 is derived for $T \to \infty$ and i.i.d. Gaussian data, neither of which holds in this setting. The descriptive use of $D_k$ in this report is deliberate, but it means the results cannot support formal inferential claims about the significance of the changepoint.

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# CONCLUSION

Post-earnings announcement drift has proven to be one of the most enduring puzzles in financial economics—documented across decades, markets, and methodologies, and resistant to the standard explanations that economic theory offers for persistent return anomalies. This report has approached the phenomenon from a variance-based, functional data perspective, asking not whether prices drift in a predictable direction after S&P 500 earnings filings, but whether the variance structure of post-announcement returns is consistent with instantaneous information absorption.

The answer, for S&P 500 companies from 2015 to 2025, is no. Day 0 variance (6.67) is more than twice the stabilized medium-term level (3.05), the $D_k$ changepoint falls at Day 2 (IT = 0.214), and the $D_k$ curve remains positive through Day 28. But the filing type comparison sharpens the finding considerably. For 10-Q filings, the uncertainty spike resolves within three to four days—a fast adjustment consistent with the discrete nature of quarterly earnings numbers. For 10-K filings, variance does not decay but instead remains elevated and rises through Day 18, with a back-loaded $D_k$ curve that turns negative from Day 3

onward. The market of 2015–2025 requires as much as three weeks to fully absorb an annual filing's information content.

This finding is more modest than the classic PEAD result. The long-run drift documented extensively in pre-2010 research—spanning 60 or more trading days—does not appear in this sample. The modern market is unambiguously faster. But faster is not the same as instantaneous, and the two-day adjustment window identified here represents a genuine and measurable deviation from the efficiency benchmark. In a world where algorithms parse financial filings in milliseconds, the persistence of even a two-day adjustment window suggests that the information embedded in SEC filings is complex enough that it resists complete mechanical extraction—that human interpretation, analyst commentary, and market dialogue are still required to fully price it.

The filing type comparison is the most practically important result in this report. 10-Q filings generate a fast, front-loaded variance spike that resolves within days—consistent with investors quickly pricing a specific quarterly number against expectations. 10-K filings generate sustained elevated variance that rises through Day 18, reflecting the greater complexity and information density of annual reports. Investors and analysts monitoring filing dates as sources of short-term price uncertainty should treat 10-K and 10-Q filing dates very differently: the quarterly filing's uncertainty window closes by Day 4, while the annual filing's window extends through most of the subsequent month.

Several directions for future work emerge naturally from this analysis. First, extending the sample to include mid-cap and small-cap stocks would test whether the patterns found here for S&P 500 constituents are representative or whether they significantly understate the adjustment window in less efficient market segments. Second, conditioning on the direction of earnings surprise—splitting the sample into positive and negative SUE quintiles—would allow a direct comparison of the variance-based results with the traditional PEAD literature and would test whether the variance concentration in the early days is asymmetric in the way that Zhang, Sijia, and co-authors found for the mean drift. Third, replicating the analysis using pre-2010 data would allow a direct temporal comparison and would enable a quantitative assessment of how much the adjustment window has actually shortened over the past 30 years. Fourth, incorporating analyst forecast revision data along the lines of Zhang's work would allow the analysis to explore whether analyst responsiveness affects the speed of variance resolution as well as the mean drift.

What this report ultimately argues is that market efficiency is better understood as a dynamic process than as a static state. The modern market processes earnings information faster than its predecessors, but the complete resolution of post-announcement uncertainty still unfolds over days rather than milliseconds. The gap between "faster" and "instantaneous" is where PEAD—in its variance-concentrated, modern form—continues to live.

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# APPENDIX A: R CODE

The following R code was used to generate all figures and tables in this report. The script is also saved as `Additional Research/pead_analysis.R`. To reproduce the results, set the working directory to `Additional Research/` before running.

```r
library(tidyverse)
library(readr)
library(dplyr)
library(lubridate)

# Load and filter to S&P 500, 2015-2025
```

```r
pead            <- vroom::vroom("../Visuals and My research/daily_filing_pct_
change.csv")
sp500_companies <- read_csv("../Visuals and My research/sp500_companies.csv")
sp500lst        <- sp500_companies$Ticker
pead            <- pead[pead$ticker %in% sp500lst, ]
pead            <- pead %>% mutate(filing_date = mdy(filing_date))
pead_2015_2025  <- pead %>%
  filter(between(filing_date, ymd("2015-01-01"), ymd("2025-12-31")))

df       <- pead_2015_2025
day_cols <- paste0("day", 0:30, "_pct_change")
dir.create("figures", showWarnings = FALSE)

# Matrix of returns: rows = filing events, cols = days 0-30
Y           <- df %>% select(all_of(day_cols)) %>% as.matrix()
colnames(Y) <- 0:30

# Estimated mean and variance functions
mu_hat  <- colMeans(Y, na.rm = TRUE)
var_hat <- apply(Y, 2, var, na.rm = TRUE)

# Inclan-Tiao cumulative sum statistics
a2      <- var_hat
T_total <- length(a2)
Ck      <- cumsum(a2)
Dk      <- Ck / Ck[T_total] - (1:T_total) / T_total
k_hat   <- which.max(abs(Dk))
day_hat <- (0:30)[k_hat]
IT_stat <- sqrt(T_total / 2) * max(abs(Dk))

# Figures 1-4: full sample (see pead_analysis.R for complete ggplot + ggsave
code)

# 10-K vs 10-Q analysis
df_10k  <- pead_2015_2025 %>% filter(filing_type == "K")
df_10q  <- pead_2015_2025 %>% filter(filing_type == "Q")
Y_10k   <- df_10k %>% select(all_of(day_cols)) %>% as.matrix()
Y_10q   <- df_10q %>% select(all_of(day_cols)) %>% as.matrix()
var_10k <- apply(Y_10k, 2, var, na.rm = TRUE)
var_10q <- apply(Y_10q, 2, var, na.rm = TRUE)

# Figures 5-6: filing type comparison (see pead_analysis.R for complete code)
```

*Full code with all figure export calls is in `Additional Research/pead_analysis.R`.*